

MCQ Quiz – Heart Murmur Detection Project

Instructions: Choose the correct option for each question.

1. What type of data is primarily used in this project?

- A. ECG signals
- B. Heart sound audio files
- C. Text reports
- D. Medical images

Answer: B

2. What is the purpose of fixing the sampling rate to 22,050 Hz?

- A. Reduce dataset size
- B. Improve UI performance
- C. Maintain uniform audio input
- D. Increase model accuracy automatically

Answer: C

3. Why is audio duration fixed to 10 seconds?

- A. To remove noise
- B. To reduce file size
- C. To ensure consistent model input
- D. To increase dataset

Answer: C

4. Which feature extraction technique is used in the project?

- A. FFT
- B. Spectrogram
- C. MFCC
- D. Wavelet Transform

Answer: C

5. What does MFCC represent?

- A. Raw audio values
- B. Image pixels
- C. Human-like sound features
- D. Heart rate values

Answer: C

6. Why are unlabelled data samples assigned label -1?

- A. For training
- B. To remove them
- C. To mark them as test-only data
- D. For balancing

Answer: C

7. Which model is used for classification?

- A. CNN
- B. Random Forest
- C. LSTM
- D. KNN

Answer: C

8. Why is LSTM suitable for this project?

- A. Handles images
- B. Works with sequences
- C. Requires less data
- D. Faster training

Answer: B

9. How is class imbalance handled in the project?

- A. Oversampling
- B. Undersampling
- C. Class weights
- D. Data removal

Answer: C

10. What framework is used to build the web application?

- A. Flask
- B. Django
- C. Streamlit
- D. FastAPI

Answer: C

11. Where is the trained model stored?

- A. Google Drive
- B. Local system
- C. Kaggle
- D. Hugging Face Hub

Answer: D

12. What file types can be uploaded in the Streamlit app?

- A. JPG, PNG
- B. CSV only
- C. WAV and MP3
- D. TXT

Answer: C

13. Why is preprocessing reused during inference?

- A. For faster prediction
- B. To avoid data mismatch
- C. To reduce memory
- D. For UI consistency

Answer: B

14. What does the model output represent?

- A. Audio signal
- B. MFCC values
- C. Class probabilities
- D. Sampling rate

Answer: C

15. What is the main goal of deploying this project?

- A. Data collection
- B. Model training
- C. User-friendly murmur detection
- D. Dataset creation

Answer: C